Dosadašnje tehnike - BoW i TFIDF

counting of words

word embeddings

pretvaranje teksta u vektore gdje se riječi sličnog značenja nalaze blizu u vektorskom prostoru

riječi koje imaju suprotno značenje se nalaze daleko od svojih protuznačnica

riječi koje imaju drugačije značenje će zadržati svoje razlike

model uči asocijacije riječi iz velikog korpusa teksta

poslije je sposoban samostalno otkriti nove sinonime ili suprotne riječi

Uvodimo tehniku pod nazivom word2vec - deep learning model

Prednosti

- smanjenje dimenzionalnosti
- razumijevanje konteksta

CBow

Skipgram

CBOW demo

ovo je samo demo, nije čitav model

```
sentence = "the king rules the kingdom"
words = sentence.split()

window_size = 1
cbow_data = []

for i in range(window_size, len(words) - window_size):
    context = [words[i - 1], words[i + 1]]
    target = words[i]
    cbow_data.append((context, target))

print("CBOW training examples:")
for context, target in cbow_data:
    print(f"Input: {context} -> Output: {target}")
```

```
CBOW training examples:
Input: ['the', 'rules'] -> Output: king
Input: ['king', 'the'] -> Output: rules
Input: ['rules', 'kingdom'] -> Output: the
```

Skip gram demo

```
sentence = "the king rules the kingdom"
words = sentence.split()
```

```
window_size = 1
skipgram_data = []

for i in range(window_size, len(words) - window_size):
    center_word = words[i]
    context_words = [words[i - 1], words[i + 1]]

    for context_word in context_words:
        skipgram_data.append((center_word, context_word))

print("Skip-gram training examples:")
for center, context in skipgram_data:
    print(f"Input: {center} -> Output: {context}")
```

```
Skip-gram training examples:
Input: king -> Output: the
Input: king -> Output: rules
Input: rules -> Output: king
Input: rules -> Output: the
Input: the -> Output: rules
Input: the -> Output: kingdom
```

U praksi se koriste gotovi modeli

```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 50.7 MB/s eta 0:00:00
```

```
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import gensim
```

gotov Python library

koristi se za treniranje svog modela

učitavanje gotovog modela

```
from gensim.models import Word2Vec
```

Word2Vec koristimo kada želimo trenirati vlastiti model i naučiti vektorske reprezentacije riječi iz našeg skupa podataka. To ima smisla kada radimo s domenama koje nisu dobro pokrivene postojećim modelima (npr. specifični poslovni ili tehnički tekstovi).

link na gotov predtrenirani model

https://huggingface.co/fse/word2vec-google-news-300

```
import gensim.downloader as api
wv = api.load ("word2vec-google-news-300")
```

```
[==================================================] 100.0% 1662.8/1662.8MB downloaded
```

```
vec_king = wv["king"]
```

```
vec_king
```

```
array([ 1.25976562e-01,  2.97851562e-02,  8.60595703e-03,  1.39648438e-01,
       -2.56347656e-02, -3.61328125e-02,  1.11816406e-01, -1.98242188e-01,
        5.12695312e-02,  3.63281250e-01, -2.42187500e-01, -3.02734375e-01,
       -1.77734375e-01, -2.49023438e-02, -1.67968750e-01, -1.69921875e-01,
        3.46679688e-02,  5.21850586e-03,  4.63867188e-02,  1.28906250e-01,
        1.36718750e-01,  1.12792969e-01,  5.95703125e-02,  1.36718750e-01,
        1.01074219e-01, -1.76757812e-01, -2.51953125e-01,  5.98144531e-02,
        3.41796875e-01, -3.11279297e-02,  1.04492188e-01,  6.17675781e-02,
        1.24511719e-01,  4.00390625e-01, -3.22265625e-01,  8.39843750e-02,
        3.90625000e-02,  5.85937500e-03,  7.03125000e-02,  1.72851562e-01,
        1.38671875e-01, -2.31445312e-01,  2.83203125e-01,  1.42578125e-01,
        3.41796875e-01, -2.39257812e-02, -1.09863281e-01,  3.32031250e-02,
       -5.46875000e-02,  1.53198242e-02, -1.62109375e-01,  1.58203125e-01,
       -2.59765625e-01,  2.01416016e-02, -1.63085938e-01,  1.35803223e-03,
       -1.44531250e-01, -5.68847656e-02,  4.29687500e-02, -2.46582031e-02,
        1.85546875e-01,  4.47265625e-01,  9.58251953e-03,  1.31835938e-01,
        9.86328125e-02, -1.85546875e-01, -1.00097656e-01, -1.33789062e-01,
       -1.25000000e-01,  2.83203125e-01,  1.23046875e-01,  5.32226562e-02,
       -1.77734375e-01,  8.59375000e-02, -2.18505859e-02,  2.05078125e-02,
       -1.39648438e-01,  2.51464844e-02,  1.38671875e-01, -1.05468750e-01,
        1.38671875e-01,  8.88671875e-02, -7.51953125e-02, -2.13623047e-02,
        1.72851562e-01,  4.63867188e-02, -2.65625000e-01,  8.91113281e-03,
        1.49414062e-01,  3.78417969e-02,  2.38281250e-01, -1.24511719e-01,
       -2.17773438e-01, -1.81640625e-01,  2.97851562e-02,  5.71289062e-02,
       -2.89306641e-02,  1.24511719e-02,  9.66796875e-02, -2.31445312e-01,
        5.81054688e-02,  6.68945312e-02,  7.08007812e-02, -3.08593750e-01,
       -2.14843750e-01,  1.45507812e-01, -4.27734375e-01, -9.39941406e-03,
        1.54296875e-01, -7.66601562e-02,  2.89062500e-01,  2.77343750e-01,
       -4.86373901e-04, -1.36718750e-01,  3.24218750e-01, -2.46093750e-01,
       -3.03649902e-03, -2.11914062e-01,  1.25000000e-01,  2.69531250e-01,
        2.04101562e-01,  8.25195312e-02, -2.01171875e-01, -1.60156250e-01,
       -3.78417969e-02, -1.20117188e-01,  1.15234375e-01, -4.10156250e-02,
       -3.95507812e-02, -8.98437500e-02,  6.34765625e-03,  2.03125000e-01,
        1.86523438e-01,  2.73437500e-01,  6.29882812e-02,  1.41601562e-01,
       -9.81445312e-02,  1.38671875e-01,  1.82617188e-01,  1.73828125e-01,
        1.73828125e-01, -2.37304688e-01,  1.78710938e-01,  6.34765625e-02,
        2.36328125e-01, -2.08984375e-01,  8.74023438e-02, -1.66015625e-01,
       -7.91015625e-02,  2.43164062e-01, -8.88671875e-02,  1.26953125e-01,
       -2.16796875e-01, -1.73828125e-01, -3.59375000e-01, -8.25195312e-02,
       -6.49414062e-02,  5.07812500e-02,  1.35742188e-01, -7.47070312e-02,
```

```
       -1.64062500e-01,  1.15356445e-02,  4.45312500e-01, -2.15820312e-01,
       -1.11328125e-01, -1.92382812e-01,  1.70898438e-01, -1.25000000e-01,
        2.65502930e-03,  1.92382812e-01, -1.74804688e-01,  1.39648438e-01,
        2.92968750e-01,  1.13281250e-01,  5.95703125e-02, -6.39648438e-02,
        9.96093750e-02, -2.72216797e-02,  1.96533203e-02,  4.27246094e-02,
       -2.46093750e-01,  6.39648438e-02, -2.25585938e-01, -1.68945312e-01,
        2.89916992e-03,  8.20312500e-02,  3.41796875e-01,  4.32128906e-02,
        1.32812500e-01,  1.42578125e-01,  7.61718750e-02,  5.98144531e-02,
       -1.19140625e-01,  2.74658203e-03, -6.29882812e-02, -2.72216797e-02,
       -4.82177734e-03, -8.20312500e-02, -2.49023438e-02, -4.00390625e-01,
       -1.06933594e-01,  4.24804688e-02,  7.76367188e-02, -1.16699219e-01,
        7.37304688e-02, -9.22851562e-02,  1.07910156e-01,  1.58203125e-01,
        4.24804688e-02,  1.26953125e-01,  3.61328125e-02,  2.67578125e-01,
       -1.01074219e-01, -3.02734375e-01, -5.76171875e-02,  5.05371094e-02,
        5.26428223e-04, -2.07031250e-01, -1.38671875e-01, -8.97216797e-03,
       -2.78320312e-02, -1.41601562e-01,  2.07031250e-01, -1.58203125e-01,
        1.27929688e-01,  1.49414062e-01, -2.24609375e-02, -8.44726562e-02,
```

```
vec_croatia = wv["Croatia"]
```

```
vec_croatia
```

```
       -0.02075195, -0.19824219,  0.02392578,  0.08935547, -0.05078125,
        0.36328125, -0.04589844, -0.09375   ,  0.04589844, -0.14550781,
        0.08837891, -0.03125   , -0.20507812,  0.00653076,  0.19628906,
       -0.11621094, -0.13671875, -0.04614258, -0.20410156,  0.14941406,
       -0.10009766, -0.1328125 ,  0.12451172,  0.15234375, -0.02355957,
        0.15039062,  0.09472656,  0.33789062,  0.00195312,  0.11523438,
       -0.12695312,  0.19824219, -0.04101562,  0.08105469,  0.31445312,
       -0.09375   ,  0.06494141,  0.00622559, -0.03564453, -0.28320312,
        0.09033203, -0.453125  , -0.19628906, -0.14160156,  0.13183594,
        0.10644531,  0.09375   , -0.15039062, -0.08007812, -0.04345703,
       -0.09228516,  0.5625    , -0.22949219, -0.08105469, -0.14453125,
       -0.18652344, -0.1328125 ,  0.17773438, -0.19042969, -0.26953125,
       -0.40234375,  0.16894531, -0.13867188, -0.00695801,  0.29101562,
       -0.2734375 , -0.23046875, -0.15234375, -0.234375  ,  0.19921875,
        0.05517578, -0.09912109,  0.02636719,  0.3984375 , -0.00148773,
```

```
        0.22851562,  0.1171875 , -0.08447266,  0.03662109, -0.10888672,
        0.21289062, -0.0625    ,  0.00180054, -0.3203125 , -0.046875  ,
        0.01586914,  0.18066406,  0.09619141,  0.44335938,  0.2578125 ,
       -0.21484375,  0.06054688, -0.02062988, -0.09570312,  0.02612305,
        0.03613281,  0.109375  , -0.00732422, -0.60546875,  0.08154297,
       -0.12109375,  0.16503906, -0.21289062, -0.22167969, -0.21289062,
       -0.18261719, -0.20410156,  0.13183594,  0.12988281, -0.16308594,
        0.13964844, -0.11083984,  0.3125    , -0.05126953,  0.19921875,
        0.04272461,  0.07324219,  0.20214844, -0.20605469,  0.20605469,
        0.34570312, -0.00247192,  0.05078125, -0.02233887,  0.24902344,
       -0.10107422, -0.19824219,  0.02319336,  0.06396484,  0.15234375,
       -0.06542969, -0.02233887,  0.00656128, -0.08300781,  0.18359375,
        0.02185059, -0.18261719,  0.16992188, -0.05395508,  0.08496094,
       -0.2109375 ,  0.03466797, -0.02380371,  0.12158203, -0.12011719,
       -0.19824219, -0.09179688, -0.0625    ,  0.05419922,  0.01135254,
        0.18554688, -0.24023438, -0.13671875, -0.06689453, -0.09521484,
        0.05322266,  0.09130859, -0.13085938, -0.12207031,  0.14648438,
       -0.109375  , -0.07861328,  0.05566406, -0.0546875 ,  0.30273438,
        0.16796875, -0.10546875,  0.0612793 , -0.3984375 , -0.35742188,
       -0.08789062, -0.04101562, -0.16308594, -0.17089844, -0.00069809,
       -0.21875   ,  0.11816406, -0.26953125, -0.21191406,  0.04541016,
       -0.15722656,  0.00817871, -0.12988281,  0.30078125, -0.19238281,
        0.03857422,  0.24316406, -0.03930664,  0.09423828, -0.24707031,
       -0.34960938, -0.17382812, -0.12792969, -0.03393555, -0.14160156,
        0.04980469,  0.3046875 ,  0.08154297,  0.24511719, -0.23730469,
       -0.1953125 , -0.07373047,  0.15527344, -0.05932617,  0.02001953,
        0.36914062, -0.06152344,  0.36132812, -0.07324219, -0.03271484,
        0.23144531,  0.0043335 ,  0.09326172,  0.25195312,  0.01965332,
        0.16503906, -0.10302734,  0.07958984,  0.06176758, -0.10498047,
        0.296875  ,  0.11865234,  0.21972656,  0.37890625,  0.05932617,
        0.31445312,  0.46484375,  0.11816406,  0.11279297,  0.07617188,
        0.37890625, -0.09716797,  0.19335938,  0.17773438,  0.47460938,
       -0.04589844, -0.20703125, -0.26367188, -0.01977539, -0.01470947,
        0.19628906,  0.23339844,  0.11816406,  0.1484375 ,  0.31445312,
       -0.13378906,  0.20605469,  0.13085938, -0.18945312,  0.01855469,
       -0.06396484,  0.1875    , -0.06494141,  0.08837891, -0.20898438,
        0.08740234,  0.03735352,  0.04248047, -0.015625  ,  0.23730469,
        0.23925781,  0.3046875 , -0.27539062, -0.01843262,  0.03149414,
        0.18652344, -0.1484375 ,  0.04882812,  0.34570312,  0.30078125],
      dtype=float32)
```

```
vec_king.shape
```

```
(300,)
```

```
wv.most_similar ("football")
```

```
[('soccer', 0.731354832649231),
 ('fooball', 0.7139959335327148),
 ('Football', 0.7124834060668945),
 ('basketball', 0.668246865272522),
 ('footbal', 0.6649289727210999),
 ('athletics', 0.6265192627906799),
 ('gridiron', 0.6191604733467102),
 ('baseball', 0.6162001490592957),
 ('footballl', 0.6069177985191345),
 ('sports', 0.5927178859710693)]
```

```
wv.most_similar("Croatia")
```

```
[('Slovenia', 0.7942927479743958),
 ('Serbia', 0.7819275259971619),
 ('Croatians', 0.7199018597602844),
 ('Zagreb', 0.7135883569717407),
 ('Serbia_Montenegro', 0.708234429359436),
 ('Croatian', 0.706842839717865),
 ('Slovakia', 0.7066327333450317),
 ('Montenegro', 0.7044665813446045),
 ('Lithuania', 0.6917349696159363),
 ('Bosnia_Herzegovina', 0.6822905540466309)]
```

```
wv.most_similar("happy")
```

```
[('glad', 0.7408890724182129),
 ('pleased', 0.6632170677185059),
 ('ecstatic', 0.6626912355422974),
 ('overjoyed', 0.6599286794662476),
 ('thrilled', 0.6514049172401428),
 ('satisfied', 0.6437949538230896),
```

```
 ('proud', 0.636042058467865),
 ('delighted', 0.627237856388092),
 ('disappointed', 0.6269949674606323),
 ('excited', 0.6247665286064148)]
```

```
wv.similarity("sport", "football")
```

```
np.float32(0.48124415)
```

```
vec = wv["king"] - wv["man"] + wv["woman"]
```

```
vec
```

```
array([ 4.29687500e-02, -1.78222656e-01, -1.29089355e-01,  1.15234375e-01,
        2.68554688e-03, -1.02294922e-01,  1.95800781e-01, -1.79504395e-01,
        1.95312500e-02,  4.09919739e-01, -3.68164062e-01, -3.96484375e-01,
       -1.56738281e-01,  1.46484375e-03, -9.30175781e-02, -1.16455078e-01,
       -5.51757812e-02, -1.07574463e-01,  7.91015625e-02,  1.98974609e-01,
        2.38525391e-01,  6.34002686e-02, -2.17285156e-02,  0.00000000e+00,
        4.72412109e-02, -2.17773438e-01, -3.44726562e-01,  6.37207031e-02,
        3.16406250e-01, -1.97631836e-01,  8.59375000e-02, -8.11767578e-02,
       -3.71093750e-02,  3.15551758e-01, -3.41796875e-01, -4.68750000e-02,
        9.76562500e-02,  8.39843750e-02, -9.71679688e-02,  5.17578125e-02,
       -5.00488281e-02, -2.20947266e-01,  2.29492188e-01,  1.26403809e-01,
        2.49023438e-01,  2.09960938e-02, -1.09863281e-01,  5.81054688e-02,
       -3.35693359e-02,  1.29577637e-01,  2.41699219e-02,  3.48129272e-02,
       -2.60009766e-01,  2.42309570e-01, -3.21777344e-01,  1.45416260e-02,
       -1.59179688e-01, -8.37402344e-02,  1.65039062e-01,  1.58691406e-03,
        3.09570312e-01,  3.16406250e-01,  7.38525391e-03,  2.41210938e-01,
        4.90722656e-02, -9.86328125e-02,  2.90527344e-02,  1.49414062e-01,
       -4.83398438e-02,  2.35595703e-01,  2.21191406e-01,  1.25488281e-01,
       -1.38671875e-01,  1.54296875e-01,  7.18994141e-02,  1.29882812e-01,
       -1.05712891e-01,  6.00585938e-02,  3.14697266e-01,  1.09619141e-01,
        8.49609375e-02,  7.71484375e-02, -2.17285156e-02,  6.11572266e-02,
       -1.89941406e-01,  2.07519531e-01, -1.63085938e-01,  1.13525391e-01,
        2.01171875e-01,  6.06689453e-02,  1.27929688e-01, -3.11279297e-01,
       -2.80151367e-01, -1.55883789e-01,  4.15039062e-02,  9.87854004e-02,
        1.69555664e-01, -3.49121094e-02,  2.08496094e-01, -9.89990234e-02,
```

```
         4.39453125e-03, -7.27539062e-02, -4.24804688e-02, -4.09179688e-01,
        -2.76367188e-01,  1.64062500e-01, -5.57617188e-01, -2.02199936e-01,
         2.12158203e-01, -9.81445312e-02,  2.30773926e-01,  2.75878906e-01,
         1.68092728e-01, -4.50439453e-02,  1.71615601e-01, -3.77075195e-01,
        -3.52478027e-03, -3.01513672e-01,  1.74224854e-01,  3.30078125e-01,
         2.00683594e-01,  1.17736816e-01, -1.37695312e-01, -1.07421875e-01,
         8.61816406e-02,  1.06445312e-01,  1.44531250e-01,  3.05175781e-03,
         1.80664062e-02,  3.73535156e-02,  7.32421875e-03,  1.32812500e-01,
         9.61914062e-02,  3.35998535e-01,  1.81152344e-01,  2.40905762e-01,
        -8.49609375e-02, -1.10107422e-01,  2.11914062e-01,  5.85937500e-03,
         1.62109375e-01, -4.15527344e-01,  1.39160156e-01,  1.01562500e-01,
         1.44531250e-01, -1.09375000e-01,  4.88281250e-02,  6.15234375e-02,
        -1.69921875e-01,  3.28369141e-02,  5.56640625e-02,  1.47460938e-01,
        -2.24609375e-02, -2.73925781e-01, -2.81982422e-01, -1.39160156e-01,
        -1.81884766e-01,  9.33532715e-02,  1.21093750e-01, -5.37109375e-03,
        -1.87500000e-01,  3.05175781e-04,  5.52734375e-01, -9.71679688e-02,
        -1.81640625e-01, -1.51855469e-01,  7.76367188e-02, -2.38281250e-01,
        -2.63977051e-02,  2.25555420e-01, -3.02734375e-01,  1.34765625e-01,
         3.23242188e-01,  1.25976562e-01,  3.51562500e-02, -2.04345703e-01,
         2.96142578e-01,  1.03149414e-01, -4.76074219e-03,  1.69189453e-01,
        -3.50585938e-01,  2.46887207e-02, -3.90502930e-01, -2.70507812e-01,
         1.85241699e-02,  1.04492188e-01,  2.84179688e-01,  1.35009766e-01,
        -5.95703125e-02,  1.88232422e-01,  8.88214111e-02,  3.24707031e-02,
        -8.98437500e-02,  5.45043945e-02,  5.65185547e-02,  1.56860352e-01,
        -9.70458984e-03, -7.08007812e-02,  5.71289062e-02, -3.08837891e-01,
        -1.91894531e-01,  4.83398438e-02,  5.22460938e-02, -1.59667969e-01,
        -4.49218750e-02, -7.37304688e-02,  5.51757812e-02,  2.12402344e-01,
         2.05322266e-01, -2.73437500e-02,  7.86132812e-02,  3.19091797e-01,
        -1.56982422e-01, -3.92822266e-01,  4.00390625e-02,  9.93652344e-02,
        -1.97372437e-02, -8.25195312e-02,  2.53906250e-02,  3.10668945e-02,
        -3.63769531e-02,  1.48925781e-02,  2.20703125e-01, -5.98144531e-02,
         6.15234375e-02, -7.14111328e-02, -4.00390625e-02, -1.03515625e-01,
```

```
wv.most_similar([vec])
```

```
[('king', 0.8449392318725586),
 ('queen', 0.7300517559051514),
 ('monarch', 0.645466148853302),
 ('princess', 0.6156251430511475),
 ('crown_prince', 0.5818676352500916),
```

```
 ('prince', 0.5777117609977722),
 ('kings', 0.5613663792610168),
 ('sultan', 0.5376775860786438),
 ('Queen_Consort', 0.5344247817993164),
 ('queens', 0.5289887189865112)]
```

```
wv.most_similar("burek")
```

```
[('cevapcici', 0.7121683359146118),
 ('dolmas', 0.6751578450202942),
 ('beef_goulash', 0.6663990020751953),
 ('sarma', 0.6660243272781372),
 ('köfte', 0.6639270186424255),
 ('lahmacun', 0.6583811044692993),
 ('plov', 0.6566954851150513),
 ('dolma', 0.6563305258750916),
 ('ajvar', 0.6533424854278564),
 ('varenyky', 0.6522648334503174)]
```

```
wv.most_similar("čevapčići")
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_4059/3239267447.py in <cell line: 0>()
----> 1 wv.most_similar("čevapčići")

 1 frames

/usr/local/lib/python3.12/dist-packages/gensim/models/keyedvectors.py in get_mean_vector(self, keys, weights,
pre_normalize, post_normalize, ignore_missing)
    516                 total_weight += abs(weights[idx])
    517             elif not ignore_missing:
--> 518                 raise KeyError(f"Key '{key}' not present in vocabulary")
    519 
    520         if total_weight > 0:

KeyError: "Key 'čevapčići' not present in vocabulary"
```

```
wv.most_similar("cevapcici")
```

```
[('burek', 0.712168276309967),
 ('sarma', 0.6980767846107483),
 ('beef_goulash', 0.6811515688896179),
 ('köfte', 0.6662662625312805),
 ('Merguez', 0.66299968957901),
 ('kofte', 0.6607834696769714),
 ('kebob', 0.6604267954826355),
 ('varenyky', 0.6562581062316895),
 ('khachapuri', 0.6547667384147644),
 ('gyro_sandwich', 0.6547386050224304)]
```

```
wv.most_similar("ljubav")
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_4059/1614072159.py in <cell line: 0>()
----> 1 wv.most_similar("ljubav")

1 frames
/usr/local/lib/python3.12/dist-packages/gensim/models/keyedvectors.py in get_mean_vector(self, keys, weights,
pre_normalize, post_normalize, ignore_missing)
    516                 total_weight += abs(weights[idx])
    517             elif not ignore_missing:
--> 518                 raise KeyError(f"Key '{key}' not present in vocabulary")
    519
    520         if total_weight > 0:

KeyError: "Key 'ljubav' not present in vocabulary"
```

```
wv.most_similar("Zagreb")
```

```
[('Belgrade', 0.7443435192108154),
 ('Podgorica', 0.7346686720848083),
 ('Croatia', 0.7135884165763855),
 ('Ljubljana', 0.7122986316680908),
```

```
 ('Skopje', 0.7006819248199463),
 ('Croatian', 0.6881119608879089),
 ('Osijek', 0.6880931854248047),
 ('Novi_Sad', 0.686374306678772),
 ('Vilnius', 0.6810470223426819),
 ('Bucharest', 0.6795935034751892)]
```

```
wv.most_similar("Sibenik")
```

```
[('Rijeka', 0.6809065937995911),
 ('Zadar', 0.6638304591178894),
 ('Varazdin', 0.6595580577850342),
 ('Trogir', 0.6556615233421326),
 ('Slavonski_Brod', 0.646776556968689),
 ('Osijek', 0.6405457854270935),
 ('Ulcinj', 0.6395336389541626),
 ('Rovinj', 0.6334735751152039),
 ('Vinkovci', 0.6188248991966248),
 ('Montenegrin', 0.6178672313690186)]
```

Počnite pisati kôd ili generirati uz pomoć umjetne inteligencije.